Q2	Create users and groups and verify least-privilege access	5 Marks

Objective: Set up user accounts and groups so each user has only the access they need, following the principle of least privilege.

Step-by-Step Algorithm

Step	Phase	Action	Command / Tool	What You Should See
1	CONFIGURE	Create a dedicated group for a team or role.	`sudo groupadd secteam`	Group in /etc/group
2	CONFIGURE	Create a user with a home directory and shell.	`sudo useradd -m -s /bin/bash analyst`	User in /etc/passwd
3	CONFIGURE	Add the user to the appropriate group.	`sudo usermod -aG secteam analyst`	Membership set
4	TEST	Check the user's group membership.	groups analyst	Groups listed
5	TEST	Confirm the user has no unintended sudo rights.	`sudo -l -U analyst`	Only intended rights

Phase key: ● SETUP · ● CONFIGURE · ● EXECUTE · ● TEST · ● ANALYSE · ● REPORT · ● FINISH

Worked Example (with sample values & output)

Applying the steps to lab-vm

A security group and an analyst account are created, then the account is checked to confirm it only belongs to the intended groups and has no administrative privileges.

User	Primary group	Extra groups	Sudo rights
analyst	analyst	secteam	**None**
admin	admin	sudo	**Full**

Sample output:

```
$ groups analyst
analyst : analyst secteam
```

Reusable Template

created 4 users(alice, bob, David, Charlie) and 3 groups (security, development, hr)

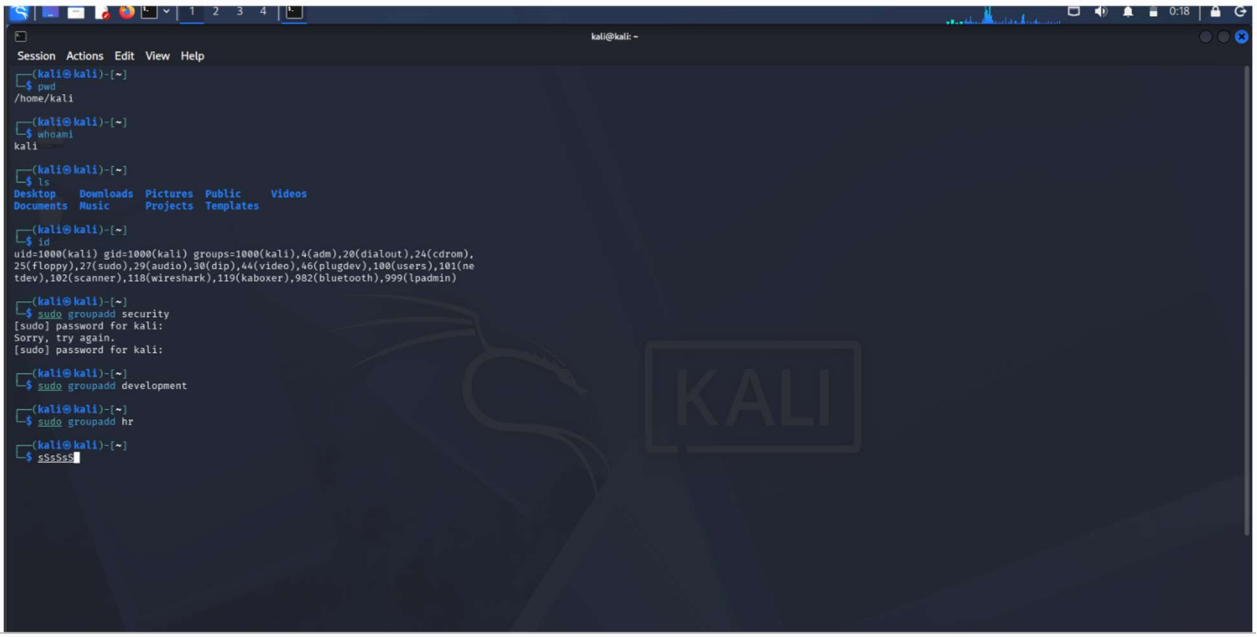
added user to group

User-access record template

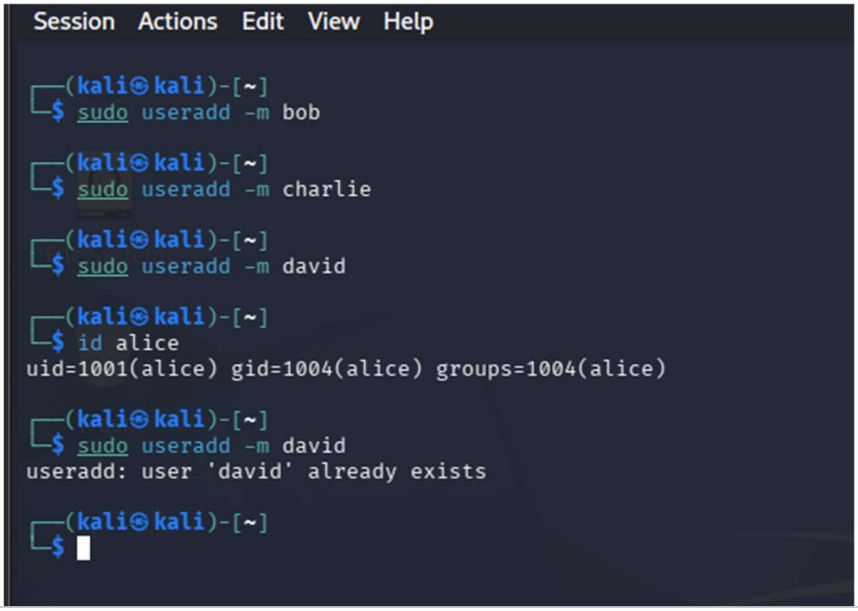
User	Groups	Least-privilege verified? (Y/N)
alice	security	N
bob	development	N
charlie	Security	N
david	hr	N

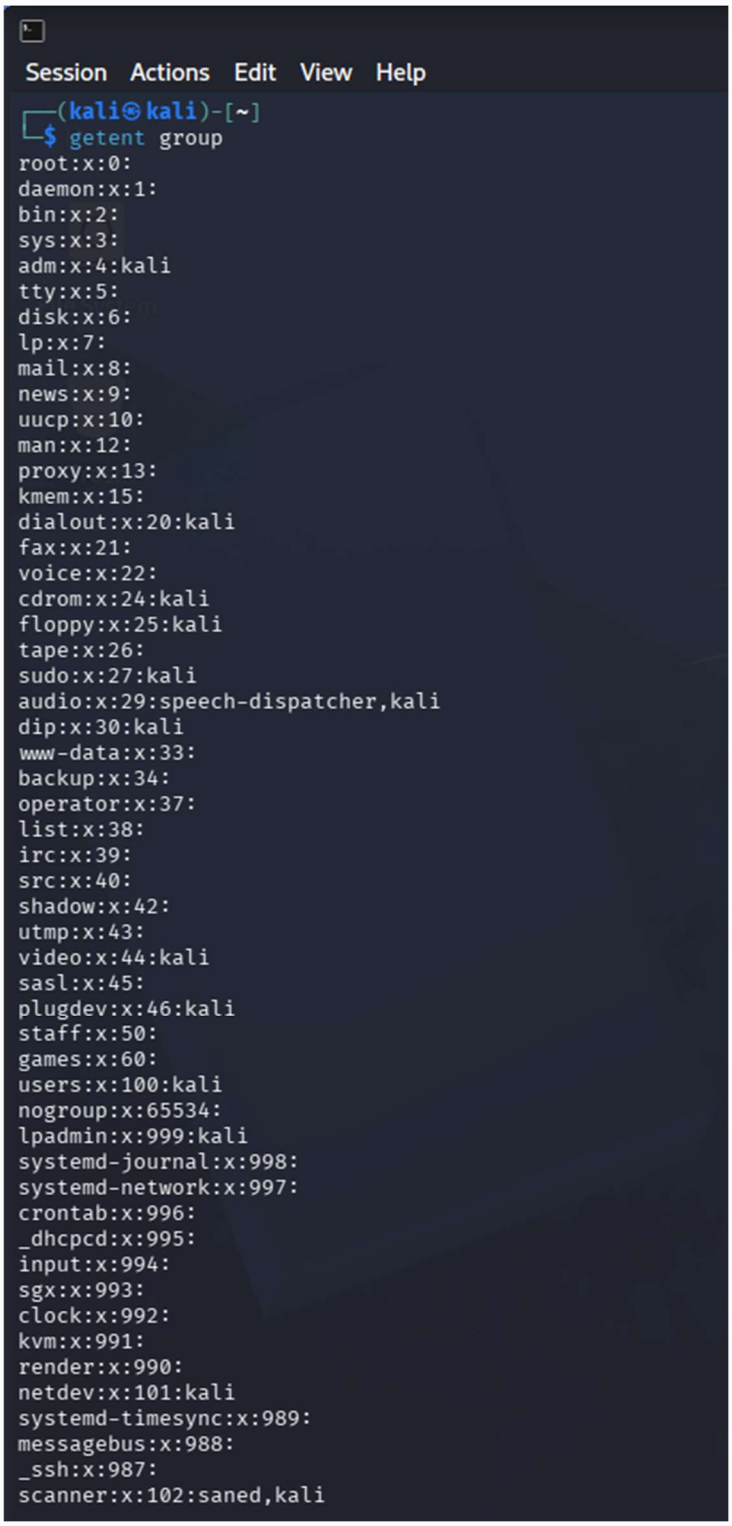
Your Output / Screenshot (paste result here):

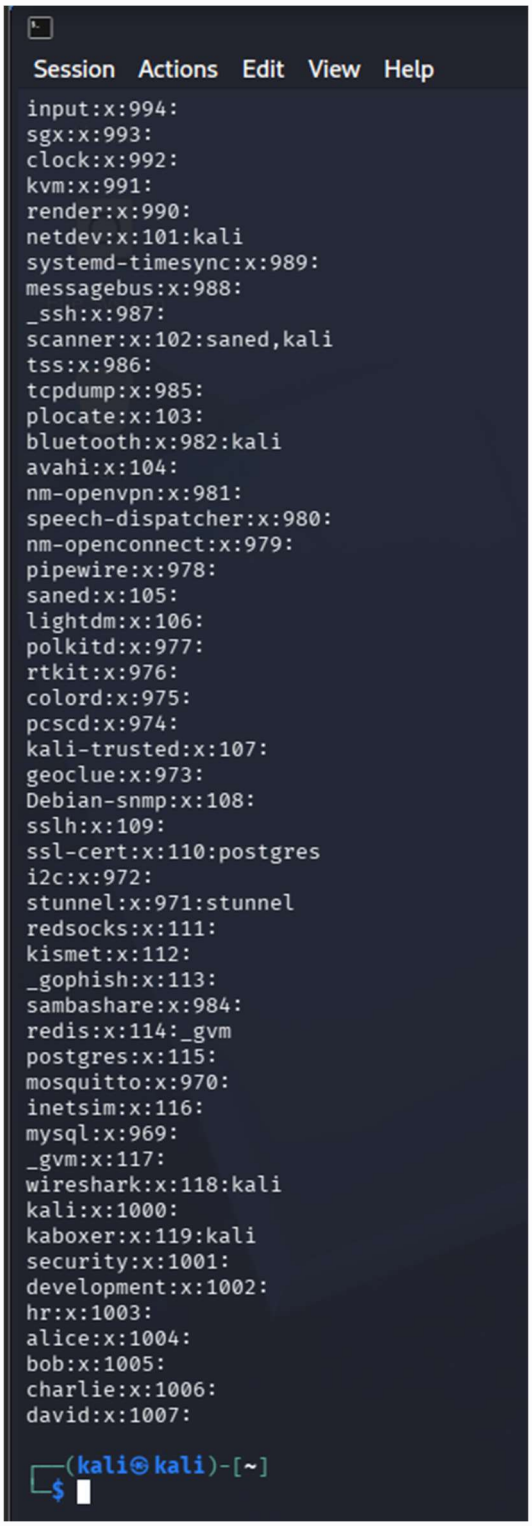
1. Created 3 groups



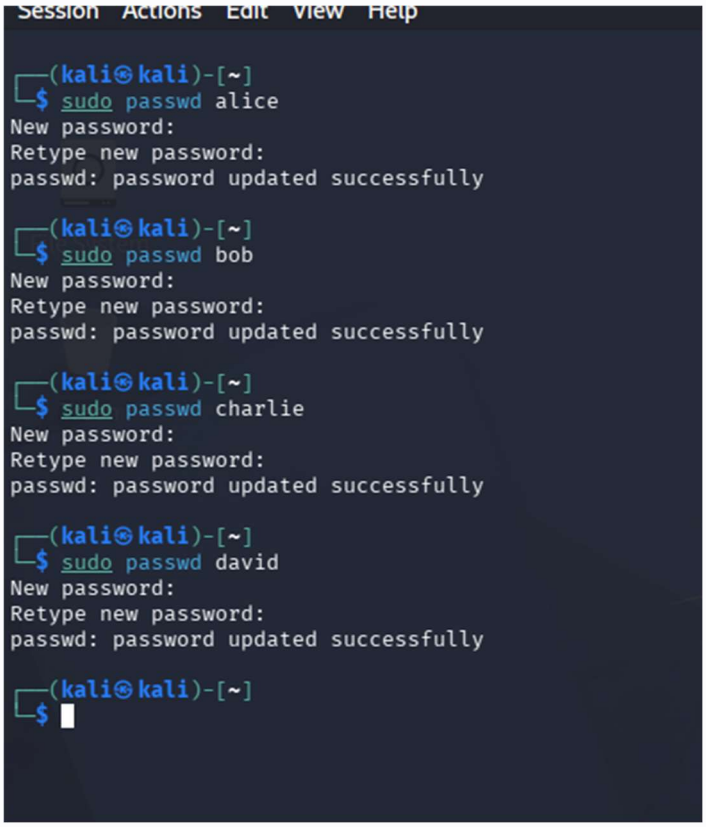
2. Created 4 users



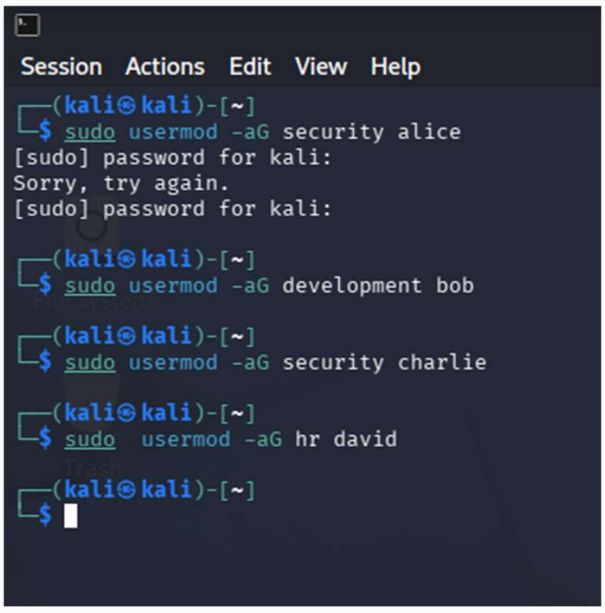

Session Actions Edit View Help
(kali㉿kali)-[~]
$ getent group
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:kali
tty:x:5:
disk:x:6:
lp:x:7:
mail:x:8:
news:x:9:
uucp:x:10:
man:x:12:
proxy:x:13:
kmem:x:15:
dialout:x:20:kali
fax:x:21:
voice:x:22:
cdrom:x:24:kali
floppy:x:25:kali
tape:x:26:
sudo:x:27:kali
audio:x:29:speech-dispatcher,kali
dip:x:30:kali
www-data:x:33:
backup:x:34:
operator:x:37:
list:x:38:
irc:x:39:
src:x:40:
shadow:x:42:
utmp:x:43:
video:x:44:kali
sasl:x:45:
plugdev:x:46:kali
staff:x:50:
games:x:60:
users:x:100:kali
nogroup:x:65534:
lpadmin:x:999:kali
systemd-journal:x:998:
systemd-network:x:997:
crontab:x:996:
_dhcpcd:x:995:
input:x:994:
sgx:x:993:
clock:x:992:
kvm:x:991:
render:x:990:
netdev:x:101:kali
systemd-timesync:x:989:
messagebus:x:988:
_ssh:x:987:
scanner:x:102:saned,kali


Session Actions Edit View Help
input:x:994:
sgx:x:993:
clock:x:992:
kvm:x:991:
render:x:990:
netdev:x:101:kali
systemd-timesync:x:989:
messagebus:x:988:
_ssh:x:987:
scanner:x:102:saned,kali
tss:x:986:
tcpdump:x:985:
plocate:x:103:
bluetooth:x:982:kali
avahi:x:104:
nm-openvpn:x:981:
speech-dispatcher:x:980:
nm-openconnect:x:979:
pipewire:x:978:
saned:x:105:
lightdm:x:106:
polkitd:x:977:
rtkit:x:976:
colord:x:975:
pcscd:x:974:
kali-trusted:x:107:
geoclue:x:973:
Debian-snmp:x:108:
sslh:x:109:
ssl-cert:x:110:postgres
i2c:x:972:
stunnel:x:971:stunnel
redsocks:x:111:
kismet:x:112:
_gophish:x:113:
sambashare:x:984:
redis:x:114:_gvm
postgres:x:115:
mosquitto:x:970:
inetsim:x:116:
mysql:x:969:
_gvm:x:117:
wireshark:x:118:kali
kali:x:1000:
kaboxer:x:119:kali
security:x:1001:
development:x:1002:
hr:x:1003:
alice:x:1004:
bob:x:1005:
charlie:x:1006:
david:x:1007:
(kali㉿kali)-[~]
$

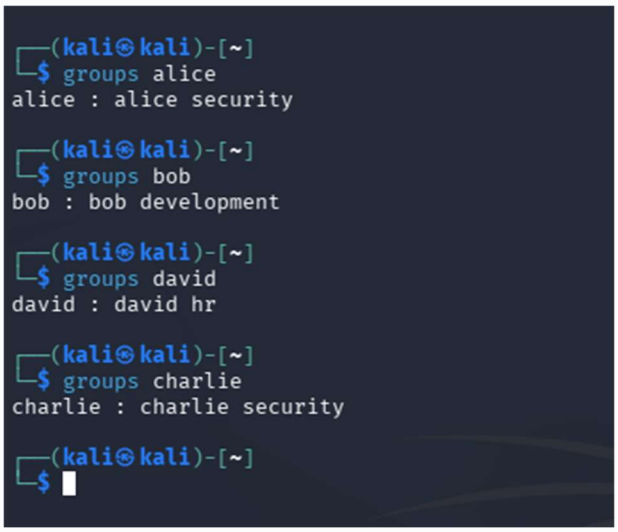
3. **Setting password to user**



4. **Adding user to groups**



. Checking whether user Is in group or not



***. Checking least priviledges**

```
┌──(kali㉿kali)-[~]
└─$ sudo  -l -u alice
usage: sudo -h | -K | -k | -V
usage: sudo -v [-ABkNnS] [-g group] [-h host] [-p prompt] [-u user]
usage: sudo -l [-ABkNnS] [-g group] [-h host] [-p prompt] [-U user]
            [-u user] [command [arg ...]]
usage: sudo [-ABbEHkNnPS] [-r role] [-t type] [-C num] [-D directory]
            [-g group] [-h host] [-p prompt] [-R directory] [-T timeout]
            [-u user] [VAR=value] [-i | -s] [command [arg ...]]
usage: sudo -e [-ABkNnS] [-r role] [-t type] [-C num] [-D directory]
            [-g group] [-h host] [-p prompt] [-R directory] [-T timeout]
            [-u user] file ...

┌──(kali㉿kali)-[~]
└─$ sudo  -l -u bob
usage: sudo -h | -K | -k | -V
usage: sudo -v [-ABkNnS] [-g group] [-h host] [-p prompt] [-u user]
usage: sudo -l [-ABkNnS] [-g group] [-h host] [-p prompt] [-U user]
            [-u user] [command [arg ...]]
usage: sudo [-ABbEHkNnPS] [-r role] [-t type] [-C num] [-D directory]
            [-g group] [-h host] [-p prompt] [-R directory] [-T timeout]
            [-u user] [VAR=value] [-i | -s] [command [arg ...]]
usage: sudo -e [-ABkNnS] [-r role] [-t type] [-C num] [-D directory]
            [-g group] [-h host] [-p prompt] [-R directory] [-T timeout]
            [-u user] file ...

┌──(kali㉿kali)-[~]
└─$ sudo  -l -u david
usage: sudo -h | -K | -k | -V
usage: sudo -v [-ABkNnS] [-g group] [-h host] [-p prompt] [-u user]
usage: sudo -l [-ABkNnS] [-g group] [-h host] [-p prompt] [-U user]
            [-u user] [command [arg ...]]
usage: sudo [-ABbEHkNnPS] [-r role] [-t type] [-C num] [-D directory]
            [-g group] [-h host] [-p prompt] [-R directory] [-T timeout]
            [-u user] [VAR=value] [-i | -s] [command [arg ...]]
usage: sudo -e [-ABkNnS] [-r role] [-t type] [-C num] [-D directory]
            [-g group] [-h host] [-p prompt] [-R directory] [-T timeout]
            [-u user] file ...

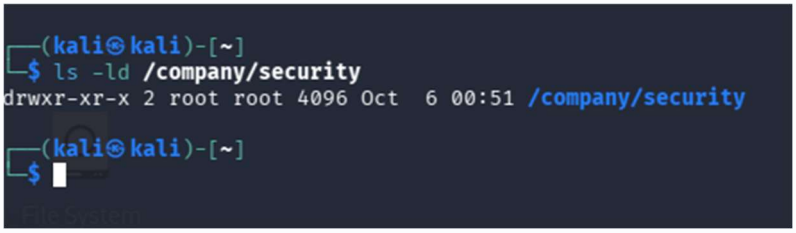
┌──(kali㉿kali)-[~]
└─$ sudo  -l -u charlie
usage: sudo -h | -K | -k | -V
usage: sudo -v [-ABkNnS] [-g group] [-h host] [-p prompt] [-u user]
usage: sudo -l [-ABkNnS] [-g group] [-h host] [-p prompt] [-U user]
            [-u user] [command [arg ...]]
usage: sudo [-ABbEHkNnPS] [-r role] [-t type] [-C num] [-D directory]
            [-g group] [-h host] [-p prompt] [-R directory] [-T timeout]
            [-u user] [VAR=value] [-i | -s] [command [arg ...]]
usage: sudo -e [-ABkNnS] [-r role] [-t type] [-C num] [-D directory]
            [-g group] [-h host] [-p prompt] [-R directory] [-T timeout]
            [-u user] file ...

┌──(kali㉿kali)-[~]
└─$
```

5. **Making a new directory and adding group to it**



6. **Checking roots**



7. **Adding text files**



8. **Adding text to thosefile**

